



Kuwait University
Faculty of Science
Department of Mathematics

MATH111: Linear Algebra

Semester: Summer 2022

Midterm 1: Sunday, July 3, 2022, 6:30pm-8pm.

Duration: 90 minutes.

Solution Key

Advice for grading computational problems. *Don't bother finding all mistakes. If it is wrong, but some parts are right, give half points. Otherwise adjust as you feel fit. Go to the first mistake and that is it. You are under no obligation to trace follow-up errors.*

Problem 1

- (a) [4pts] What is a linear system? How can you solve a linear system?

A linear system is a system of linear equations such as

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m \end{aligned}$$

To solve it, we form the coefficient matrix (possibly extended by the b_i), and apply RREF algorithm to transform it into row-echelon form. From there, we can read off the solutions.

- (b) [4pts] What is a matrix? Make an example.

A matrix is a rectangular array of numbers. We can write $A = (a_{ij})$, where a_{ij} is the entry in row i and column j of A .

Example: $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}.$

(c) [8pts] Find the inverse of $A = \begin{bmatrix} 1 & -2 & 0 \\ 2 & 0 & 1 \\ -1 & 1 & 0 \end{bmatrix}$.

$$\begin{aligned} &\text{Apply RREF to } \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 2 & 0 & 1 & 0 & 1 & 0 \\ -1 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \xrightarrow{-2R_1 + R_2 \rightarrow R_2 \text{ and } R_1 + R_3 \rightarrow R_3} \\ &\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 4 & 1 & -2 & 1 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \end{array} \right] \xrightarrow{R_2 \leftrightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 & 1 \\ 0 & 4 & 1 & -2 & 1 & 0 \end{array} \right] \xrightarrow{-R_2 \rightarrow R_2} \\ &\left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 4 & 1 & -2 & 1 & 0 \end{array} \right] \xrightarrow{-4R_2 + R_3 \rightarrow R_3} \left[\begin{array}{ccc|ccc} 1 & -2 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 & 1 & 4 \end{array} \right] \xrightarrow{2R_2 + R_1 \rightarrow R_2} \\ &\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -1 & 0 & -2 \\ 0 & 1 & 0 & -1 & 0 & -1 \\ 0 & 0 & 1 & 2 & 1 & 4 \end{array} \right] \text{ So } A^{-1} = \begin{bmatrix} -1 & 0 & -2 \\ -1 & 0 & -1 \\ 2 & 1 & 4 \end{bmatrix} \end{aligned}$$

(d) [4pts] Given a matrix

$$A = \begin{bmatrix} 5 & -3 \\ 2 & x \end{bmatrix},$$

what should x be so that A is singular?

$A = \begin{bmatrix} 5 & -3 \\ 2 & x \end{bmatrix}$ is singular if and only if $\det(A) = 5x - 2(-3) = 0$ which is iff $5x = -6$ which is iff $x = -\frac{6}{5}$.

Second way: $\left[\begin{array}{cc|cc} 5 & -3 & 1 & 0 \\ 2 & x & 0 & 1 \end{array} \right] \Leftrightarrow \left[\begin{array}{cc|cc} 1 & \frac{-3}{5} & \frac{1}{5} & 0 \\ 2 & x & 0 & 1 \end{array} \right] \Leftrightarrow \left[\begin{array}{cc|cc} 1 & \frac{-3}{5} & \frac{1}{5} & 0 \\ 0 & x + \frac{6}{5} & \frac{\frac{1}{5}}{5} & 1 \end{array} \right]$ The system is not invertible iff the last leading is zero iff $x + \frac{6}{5} = 0$ which is iff $x = -\frac{6}{5}$.

Problem 2

Define the following matrices and vectors:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, \quad B = \begin{bmatrix} 6 & 5 \\ 4 & 3 \\ 2 & 1 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} -1 \\ -2 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix}.$$

Compute the following expressions, if possible:

(a) [3pts] AB

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} 6 & 5 \\ 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6+8+6 & 5+6+3 \\ 24+20+12 & 20+15+6 \end{bmatrix} = \begin{bmatrix} 20 & 14 \\ 56 & 41 \end{bmatrix}.$$

(b) [3pts] BA

$$BA = \begin{bmatrix} 6 & 5 \\ 4 & 3 \\ 2 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 6+20 & 12+25 & 18+30 \\ 4+12 & 8+15 & 12+18 \\ 2+4 & 4+5 & 6+6 \end{bmatrix} = \begin{bmatrix} 26 & 37 & 48 \\ 16 & 23 & 30 \\ 6 & 9 & 12 \end{bmatrix}$$

$$(c) \text{ [3pts] } A\mathbf{y} = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix} = \begin{bmatrix} -3 - 8 - 15 \\ -12 - 20 - 30 \end{bmatrix} = \begin{bmatrix} -26 \\ -62 \end{bmatrix}$$

$$(d) \text{ [3pts] } B\mathbf{x} = \begin{bmatrix} 6 & 5 \\ 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} = \begin{bmatrix} -6 - 10 \\ -4 - 6 \\ -2 - 2 \end{bmatrix} = \begin{bmatrix} -16 \\ -10 \\ -4 \end{bmatrix}$$

$$(e) \text{ [3pts] } \mathbf{xy}^\top = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \cdot \begin{bmatrix} -3 & -4 & -5 \end{bmatrix} = \begin{bmatrix} 3 & 4 & 5 \\ 6 & 8 & 10 \end{bmatrix}$$

$$(f) \text{ [3pts] } \mathbf{y}^\top \mathbf{y} = \begin{bmatrix} -3 & -4 & -5 \end{bmatrix} \cdot \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix} = 9 + 16 + 25 = 50.$$

$$(g) \text{ [2pts] } \mathbf{x}^\top A\mathbf{y} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} -3 \\ -4 \\ -5 \end{bmatrix} = \begin{bmatrix} -1 \\ -2 \end{bmatrix} \begin{bmatrix} -26 \\ -62 \end{bmatrix} = \begin{bmatrix} 26 + 124 \end{bmatrix} = \begin{bmatrix} 150 \end{bmatrix}$$

Problem 3

(a) [10pts] Determine all values of a , if any, for which the system

$$\begin{array}{rcl} x & +y & -az = 2 \\ 3x & +y & +3az = 5 \end{array}$$

has a solution, no solution, or infinitely many solutions.

Rewrite the system as matrix and perform RREF: $\left[\begin{array}{ccc|c} 1 & 1 & -a & 2 \\ 3 & 1 & 3a & 5 \end{array} \right] \iff \left[\begin{array}{ccc|c} 1 & 1 & -a & 2 \\ 0 & -2 & 6a & -1 \end{array} \right]$
 $\iff \left[\begin{array}{ccc|c} 1 & 1 & -a & 2 \\ 0 & 1 & -3a & \frac{1}{2} \end{array} \right] \iff \left[\begin{array}{ccc|c} 1 & 0 & 2a & \frac{3}{2} \\ 0 & 1 & -3a & \frac{1}{2} \end{array} \right]$. The system has IMS for all values of a .

(b) [10pts] Given that $C = \begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix}$, which value of n has to be chosen in the polynomial

$$f(x) = x^2 + 4x + n$$

so that

$$f(C) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}?$$

Precompute the following matrices:

$$C^2 = \begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix} \begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix} = \begin{bmatrix} 9 + 25 & 15 - 35 \\ 15 - 35 & 25 + 49 \end{bmatrix} = \begin{bmatrix} 34 & -20 \\ -20 & 74 \end{bmatrix} \text{ and } 4C = 4 \begin{bmatrix} 3 & 5 \\ 5 & -7 \end{bmatrix} = \begin{bmatrix} 12 & 20 \\ 20 & -28 \end{bmatrix} \text{ and } nI = \begin{bmatrix} n & 0 \\ 0 & n \end{bmatrix}.$$

Therefore $f(C) = C^2 + 4C + nI = \begin{bmatrix} 34 & -20 \\ -20 & 74 \end{bmatrix} + \begin{bmatrix} 12 & 20 \\ 20 & -28 \end{bmatrix} + \begin{bmatrix} n & 0 \\ 0 & n \end{bmatrix} = \begin{bmatrix} 46+n & 0 \\ 0 & 46+n \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, which means $n = -46$.

Problem 4

- (a) [16pts] Determine a, b , and c so that

$$A = \begin{bmatrix} a-b+c & 3c & 5+a \\ a & 0 & -a \\ -1-b & a & a+3c \end{bmatrix}$$

is skew symmetric.

Comparing coefficients in

$$A = \begin{bmatrix} a-b+c & 3c & 5+a \\ a & 0 & -a \\ -1-b & a & a+3c \end{bmatrix} = -A^T = \begin{bmatrix} -(a-b+c) & -a & 1+b \\ -3c & 0 & -a \\ -5-a & a & -(a+3c) \end{bmatrix}$$

leads to the system

$$\begin{aligned} a-b+c &= 0 & a-b+c &= 0 & \begin{bmatrix} 1 & -1 & 1 & | & 0 \\ 1 & 0 & 3 & | & 0 \\ 1 & -1 & 0 & | & -4 \end{bmatrix} \\ a &= -3c & \text{or } a &+3c &= 0 & \text{or } \\ 5+a &= 1+b & a-b &= -4 & \end{aligned} \quad \text{We apply RREF:}$$

$$\begin{bmatrix} 1 & 0 & 3 & | & 0 \\ 1 & -1 & 1 & | & 0 \\ 1 & -1 & 0 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 & | & 0 \\ 0 & -1 & -2 & | & 0 \\ 0 & -1 & -3 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 & | & 0 \\ 0 & 1 & 2 & | & 0 \\ 0 & 0 & -1 & | & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 3 & | & 0 \\ 0 & 1 & 2 & | & 0 \\ 0 & 0 & 1 & | & 4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 & | & -12 \\ 0 & 1 & 0 & | & -8 \\ 0 & 0 & 1 & | & 4 \end{bmatrix}$$

So, $a = -12$, $b = -8$, $c = 4$.

(b) We check: $A = \begin{bmatrix} -12 - (-8) + 4 & 3 \cdot 4 & 5 + (-12) \\ -12 & 0 & -(-12) \\ -1 - (-8) & -12 & (-12) + 3 \cdot 4 \end{bmatrix} = \begin{bmatrix} 0 & 12 & -7 \\ -12 & 0 & 12 \\ 7 & -12 & 0 \end{bmatrix} = -A^T$.

Problem 5

- (a) [10pts] Let A and B be $n \times n$ nonsingular matrices. Prove that

$$(AB)^{-1} = B^{-1}A^{-1}.$$

We check that

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = AA^{-1} = I$$

Therefore, by the uniqueness of the inverse, we have $(AB)^{-1} = B^{-1}A^{-1}$.

- (b) [10pts] Prove that if the linear system $A\mathbf{x} = \mathbf{b}$ has two different solutions $\mathbf{x}_1$ and $\mathbf{x}_2$, then it has infinitely many solutions.

We assume that $A\mathbf{x}_1 = \mathbf{b}$ and $A\mathbf{x}_2 = \mathbf{b}$. For any integer n we consider $A\left(\frac{1}{n}\mathbf{x}_1 + \frac{n-1}{n}\mathbf{x}_2\right) = A\frac{1}{n}\mathbf{x}_1 + A\frac{n-1}{n}\mathbf{x}_2 = \frac{1}{n}A\mathbf{x}_1 + \frac{n-1}{n}A\mathbf{x}_2 = \frac{1}{n}\mathbf{b} + \frac{n-1}{n}\mathbf{b} = \mathbf{b}$. Therefore, we have found infinitely many solutions. Observe that $\frac{1}{n}\mathbf{x}_1 + \frac{n-1}{n}\mathbf{x}_2$ are distinct solutions for each n .